Mask-Reconstruction-Based Decoupled Convolution Network for Hyperspectral Imagery Classification

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***Abstract*— Deep learning has attracted much attention in hyperspectral imagery (HSI) classification. However, most deep learning methods ignore the information loss during spatial–spectral feature extraction, which potentially affects the classification performance. In this article, mask-reconstruction-based decoupled convolution network (MrDCN) is proposed, which includes the decoupled feature extraction module (DFEM) to extract spectral information and spatial information of the target HSI patch, respectively. The reconstruction modules are designed to maintain the feature extraction ability of DFEM and ensure that discriminative information in high-dimensional and low-dimensional features is preserved. MrDCN outperforms the state-of-the-art methods in classification on three datasets of various scenarios, which indicates its effectiveness, and experiments on embedded devices are executed to affirm the efficiency of MrDCN.**

***Index Terms*— Central attention, decoupled feature extraction, hyperspectral image, mask reconstruction.**

I. Introduction

HYPERSPECTRAL imagery (HSI), which occupies dozens or hundreds of contiguous bands and records reflectance spectrographic information of the ground, has a wealth of discriminative data for classifying land use and land cover [1], [2], [3], [4], [5], [6], [7], [8], [9]. Deep learning methods extract more details of discriminative characteristics than the conventional feature extraction techniques. Convolutional neutral network with deep pixel-pair features (CNN-PPF) [10] learned the pixel-pair features by inputting the pairs of pixels. This network has the advantage that training samples are greatly increased, but spatial information is not used. A interleaving perception CNN (IP-CNN) [11] was suggested to incorporate heterogeneous information that improved the classification performance. In [12], the original spectral–spatial features were first subjected to randomized principal component analysis (RPCA), and a simple network of CNN (RPCACNN) extracts more details. In spectral–spatial attention networks (SSCNNs) [13], a CNN is designed to extract spatial and spectral information. Contextual CNN (CDCNN) used a multiscale CNN to extract spectral–spatial information on a patch [14]. Diverse region CNN (DRCNN) inputted the target patch and many other patches around the target pixel at the same time [15] to gain interactional contextual information. A 3-D CNN framework was suggested [16] to completely use the local spectral structure information for improving the categorization of tree species. A transfer network of topological and semantic (TSTnet) [17] is proposed to characterize topological relationships for improving robustness of classification. Nonlinear geodesic sparse hypergraph (NGSH) is proposed to use the multivariate sparse correlations of different manifold neighborhoods [18], and local-constraint-based sparse manifold hypergraph learning (LCBSMHL) is developed to fuse local constraint and manifold reconstruction [4]. Rotation-invariant attention network (RIAN) proposed a center spectral attention to avoid the influence of other categories of pixels [19]. An efficient multitask learning network (MTLN) is developed to incorporate complementary information among pixel-level classification tasks [20]. Language-aware domain generalization network (LDGnet) is proposed to learn cross-domain-invariant representation from cross-domain shared prior knowledge [21]. A grouped subspace linear semantic alignment (G-SLSA) algorithm is designed to align the source and target domains [22]. Single-source domain expansion network (SDEnet) is developed to ensure the reliability and effectiveness of domain extension [23]. Semisupervised stacked autoencoders (Semi-SAEs) proposed two SAEs to improve the generalization performance [24]. An active learning-based Siamese network (ALSN) is proposed to solve the limited labeled samples problem [25]. A spectral–spatial Siamese network (S3Net) is developed to reduce overfitting [26]. A spectral–spatial Siamese neural network with bag-of-features (S3BoF) is designed to enhance model interpretability and feature representation [27]. A differentiated-scale restricted GCN (DSR-GCN) is developed to capture refined spatial structure [28].

Considering the limited computing resources of the mobile terminal, the lightweight network is committed to ensuring the effectiveness of the algorithm while minimizing the network algorithm. A lightweight CNN (LWCNN) [29], which uses spectral–spatial Schroedinger eigenmaps to extract the spectral and spatial features, and inputs the features to especially CNN, was offered as a solution to the short sample set problem. A multiscale CNN (MS-CNNs) [30] was suggested to extract

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multiscale information for relieving the problem of limited training examples. For HSI classification, a fast patch-free global learning (FPGA) framework was presented to lower the computational cost of the patch-based learning framework. By combining the original information and more abstract hierarchical features, information compensation was developed to gain low-level features and semantic features [31], [32].

Transformer [33], [34], which takes self-attention as core, has recently found success in a variety of other machine learning applications. A central attention network (CAN) [35] is proposed to combine location information and spectral information for improving classification. For HSI classification, a bidirectional encoder (BERTHSI) [36] in Transformer was adopted. It inputs a pixel sequence including the target pixel and its surrounding pixels into the embedding module with positional information, and the results of the embedding module were inputted into the BERT module for gaining the classification feature. The attention algorithm was combined with CNN to extract the multilevel feature network (MFN-SAM) [37]. A graph information aggregation framework (GiaCFSL) [38] in cross-domain few-shot learning is developed to improve the classification performance using graph information aggregation. An attention-based second-order pooling network (A-SPN) is proposed to improve the optimization of complex hyperparameters [39]. A local transformer with spatial partition restore network (SPRLT-Net) is developed to capture global context dependencies [40]. For a variety of other techniques in deep learning, refer the paper [41].

Although the current deep learning algorithms work reasonably well for HSI classification, there are several issues. On one hand, some CNN-based or CNN-mixed approaches, including CNNHSI [42] and CNNPPF [10], only extract the spatial domain and spectral domain feature together like visible image and do not use spectral and spatial information to their maximum potential, respectively. On the other side, SSCNN [13], CDCNN [14], DRCNN [15], 3DCNN [16], confident learning-based domain adaptation (CLDA) [43], and graph convolutional networks (FusCNNGCN) [44] focus on specific classification tasks, but do not consider the overfitting problem caused by losing structure information in feature extraction.

The spatial information reconstruction module (RESPA) and spectral information reconstruction module (RESPE) implement the reconstruction of the original pixel information based on mean square error (MSE) loss. Specifically, compared with MAE masked on the original pixel, for the extracted feature information, this work's mask operation is carried out on the feature blocks containing spatial information and the feature vectors of interspectral information, respectively. In addition to the attention mechanism, prior information of neighboring correlation is also introduced into the feature learning process. To be specific, the spatial information is weighted and reconstructed by CNN and attention simultaneously, and the interspectral information is reconstructed by full connection. The function of mask operation is developed to increase the difficulty of reconstruction, so that the network can better maintain the low-dimensional original information of data, reducing the overfitting risk of the whole model. Different from the existing methods with both the attention and MAE settings, the proposed method sets them as the plug-and-play modules, and these modules can be removed after the training stage, reducing the computational complexity of the testing process.

The main contributions of this work are summarized as follows.

1) Decoupled feature extraction module (DFEM) is developed to separate the extraction of spatial information and spectrum information, which strengthens the representation of the feature of the network, and the structure is lightweight.
2) RESPA and RESPE are developed to reconstruct the target spectrum, which ensures that the discriminative information from low dimension to high dimension is retained. Hence, the overfitting phenomenon is inhibited.
3) The removable modules (RESPA and RESPE) are also designed for lightweight deployment. Experiments on embedded devices are carried out to verify its effectiveness, which proves that the design of the regularization term is beneficial for realizing the lightweight deployment.

The remainder of this article is organized as follows. Section II explicitly explains the proposed mask-reconstruction-based decoupled convolution network (MrDCN). Section III displays the experimental results to demonstrate the superior performance of MrDCN. In Section IV, we conclude this article.

II. Proposed MrDCN

The proposed MrDCN is illustrated in Fig. 1 which includes DFEM, RESPA, RESPE, and center attention classification (CAC) head. Fig. 2 shows the derivation process of MrDCN.

A. Decoupled Feature Extraction Module

In the DFEM, first, a 3-D tensor including all the pixels of input patch is mapped into a new 3-D tensor along the channel direction that extracts the features among the spectra. This tensor is used for shift operation and attention operation to extract spatial features.

1) Decouple the CNN for Feature Extraction: Our particular CNN module is called "decoupled CNN" as shown in Fig. 3. For traditional convolution that has the kernel $K_i \in R^{C_i n \times C_{\text{out}} \times k \times k}$, where C_{in} and C_{out} are the input and output channel size, respectively, and k is the kernel size. Suppose $\mathbf{F} \in R^{C_{\text{in}} \times H \times W}$ and $\mathbf{G} \in R^{C_{\text{out}} \times H \times W}$ as the input and output tensors, and H and W are the height and width, respectively, $\mathbf{f}_{ij} \in R^{C_{\text{in}}}$ and $\mathbf{g}_{ij} \in R^{C_{\text{out}}}$ are the vectors of pixel (i, j) corresponding to $\mathbf{F}$ and $\mathbf{G}$, respectively. The traditional convolution is defined as

$$\mathbf{g}_{i,j} = \sum_{p,q} \mathbf{K}_{p,q} \mathbf{f}_{i+p-\frac{k}{2}, j+q-\frac{k}{2}} \tag{1}$$

where $\mathbf{K}_{p,q} \in R^{C_{\text{out}} \times C_{\text{in}}}$, $p, q \in \{0, 1, \ldots, k-1\}$ indicates the kernel weights in relation to the kernel position indices (p, q).

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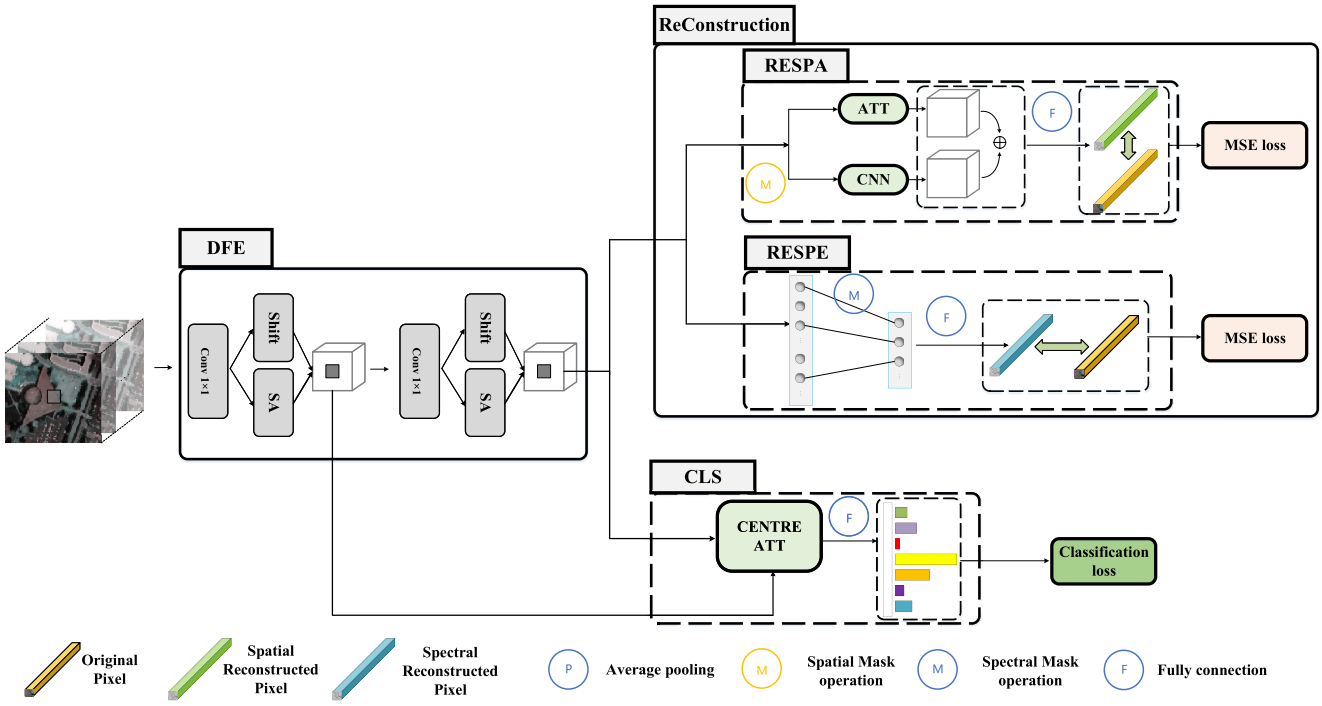


Fig. 1. Illustration for the proposed MrDCN, which includes DFEM, central attention module, RESPA, and RESPE. Shift is the shift operation in (6), and SA is the self-attention operation. DFE is the decoupled feature extraction module, CLS is the classification module, RESPA is the spatial information reconstruction module, and RESPE is the spectral information reconstruction module.

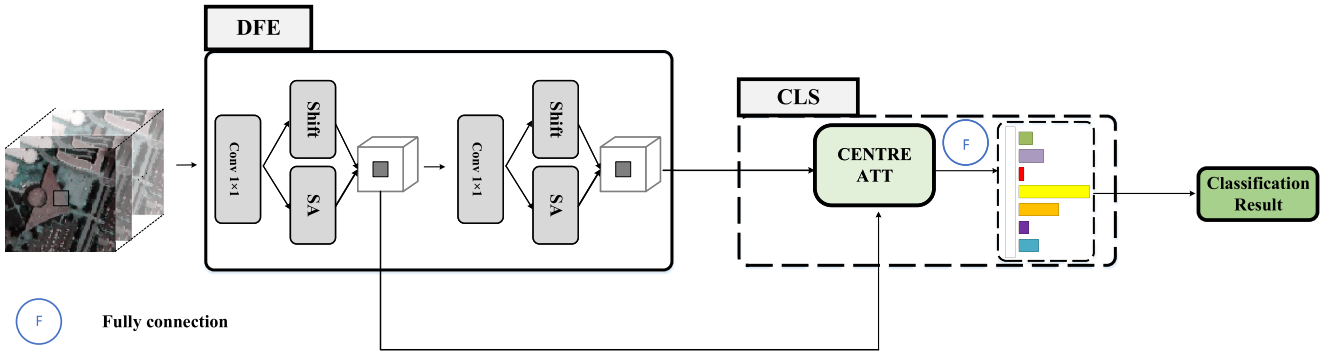


Fig. 2. Illustration for the influence process of the proposed MrDCN.

For decoupled processing, we rewrite (1) as follows:

$$\mathbf{g}_{i,j} = \sum_{p,q} \mathbf{g}_{i,j}^{(p,q)} \tag{2}$$

with

$$\sum_{p,q} \mathbf{g}_{i,j}^{(p,q)} = \mathbf{K}_{p,q}\mathbf{f}_{i+p-\frac{k}{2},j+q-\frac{k}{2}}. \tag{3}$$

Then, we can define the spatial feature extraction as shift operation

$$\hat{f} := \text{Shift}(f, x, y) = \mathbf{f}_{i+\Delta x, j+\Delta y} \quad \forall i, j \tag{4}$$

where $\Delta x and \Delta y$ represent the horizontal and vertical offsets, respectively. Then, the decoupled version convolution can be defined. We define the spectral feature extraction as

$$\hat{\mathbf{g}}_{i,j}^{(p,q)} = \mathbf{K}_{p,q}\mathbf{f}_{i,j}. \tag{5}$$

And the spatial feature extraction as

$$\mathbf{g}_{i,j} = \sum \text{Shift}\left(\hat{\mathbf{g}}_{i,j}^{(p,q)}, p - \frac{k}{2}, q - \frac{k}{2}\right). \tag{6}$$

2) Decouple the CNN With Self-Attention: As shown in Fig. 3, self-attention is a naturally decoupled feature extraction method. We take a normal self-attention block as an example. The input and output tensors are $\mathbf{F} \in R^{C_{\text{in}} \times H \times W}$, $\mathbf{G} \in R^{C_{\text{out}} \times H \times W}$, and it has N heads. Then, the spectral feature extraction of attention can be represented as

$$\mathbf{y}_{i,j} = \mathbf{ReLU}\left(\mathbf{W}_o\mathbf{x}_{i,j} + b_o\right) \tag{7}$$

where $\mathbf{x}_{i,j}$ is the input feature of pixel (i, j), $\mathbf{W}_o$ is the projection matrix for queries, RELU is the activate function, and $\mathbf{y}_{i,j}$ is the intermediate feature. The spatial feature extraction

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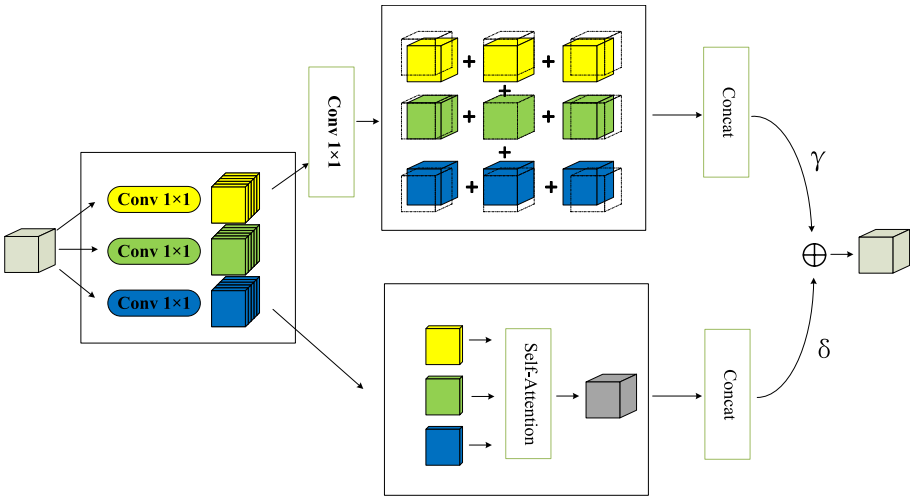


Fig. 3. Illustration for DFEM. $\gamma and \delta$ are the coefficients of weights.

of attention can be represented as

$$\mathbf{A} = \mathbf{SoftMax}\left(\frac{\mathbf{y}_{i,j}^T \mathbf{y}}{\sqrt{\text{length}(\mathbf{y}_0)}}\right) \tag{8}$$

where SoftMax is the activate function, and $\mathbf{A}$ is the attention weight which represents the relationship between the pixel (i, j) and other pixels. Then, the attention can be represented as

$$\mathbf{g}_{ij} = \mathbf{Concat}_{1,\ldots,N}\left(\sum_{p,q} \mathbf{A}\mathbf{y}_{p,q}\right) \tag{9}$$

where pixel (p, q) belongs to the neighborhood of pixel(i, j), and N is the number of heads. The attention module can share spectral feature extraction with the common 1×1 convolution. The results of attention and convolution are added by weighting coefficients. $\gamma and \delta$ are the learnable weighting coefficients. Then, this module can use the convolution and attention synchronous with lesser parameters.

B. Center Attention Classification

The input is a 3-D tensor. The height and width are $m and n$, respectively, the channel is C_i, and $m = n$. The center pixel is the target pixel of the input patch that serves as the query pixel for classification. It should be noted that $m and n$ are odd numbers. We transfer the 3-D input into a 2-D matrix $\mathbf{x}_i \in R^{C_i \times mn}$. $\mathbf{y} \in R^{C_o \times mn}$ as

$$\mathbf{y} = \mathbf{ReLU}(\mathbf{W}_o \mathbf{x}_i + \mathbf{b}_o) \tag{10}$$

where $\mathbf{W}_o \in R^{C_o \times C_i}$ and $\mathbf{b}_o \in R^{C_o \times mn}$ are the network parameters, and C_o is the channel size of $\mathbf{y}$. Then, $\mathbf{y}$ is converted into the value tensor $\mathbf{Y} \in R^{C_o \times m \times n}$. A key tensor's related key matrix is

$$\mathbf{z} = \mathbf{W}_1 \mathbf{x}_i + \mathbf{b}_1. \tag{11}$$

$R^{C_o \times mn}$, $\mathbf{W}_1 \in R^{C_o \times C_i}$, and $\mathbf{b}_1 \in R^{C_o \times mn}$.

The center vector $\mathbf{z}_0$, i.e., the key of the target pixel. A 1-D vector is

$$\mathbf{w} = \mathbf{SoftMax}\left(\frac{\mathbf{z}_o^T \mathbf{z}}{\sqrt{C_o}}\right). \tag{12}$$

The self-attention scaling factor in [33] is analogous to that of $(1/\sqrt{C_o})$. $\mathbf{w}$ is converted into a weight matrix $\mathbf{W} \in R^{m \times n}$. The output $\mathbf{X}_o \in R^{C_o \times (m-2) \times (n-2)}$ is

$$\mathbf{X}_o = \mathbf{AvgPool}(\mathbf{Y} \otimes \mathbf{W})/\mathbf{AvgPool}(\mathbf{W}) \tag{13}$$

where $\mathbf{W}$ is repeated C_0 times to adjust the size of $\mathbf{Y}$ along the channel direction, and **AvgPool** means average pooling. $\otimes$ and / are pointwise multiplication and division, respectively.

1) Multihead Central Attention: Multihead central attention is created on the basis that using several heads of attention in the Transformer is advantageous. Let the head count be h. The value tensors $Y \in R^{C_o \times m \times n}$ and $\mathbf{z} \in R^{C_o \times mn}$ are separated into $Y_1, Y_2, \ldots, Y_h \in R^{(C_o/h) \times m \times n}$ along the channel direction and $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_h \in R^{(C_o/h) \times mn}$. By (11) and (12), X_{oj} corresponding to $(Y_j, \mathbf{z}_j)$ in (13) for each j is obtained

$$X_o = \mathbf{Concat}(X_{o1}, \ldots, X_{oh}) \tag{14}$$

where **Concat** denotes the concatenation of all the tensors along the channel direction. The multihead central attention is adapted from the central attention. The advantages are dual. First, unlike self-attention, it extracts positive features without the use of an additional layer; hence, multihead attention does not require an additional 1-D convolution. Second, adding a further 1-D convolution requires training additional parameters. The complexity is thereby decreased without the additional 1-D convolution. The cross-entropy loss in classification is defined as

$$\mathcal{L}_{\text{cls}} = -\sum_{c=1}^{C} e_i^c \log p_i^c \tag{15}$$

where e_i^c is the groud truth, and p_i^c is the final result.

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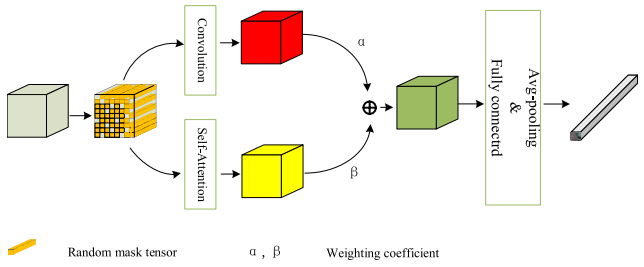


Fig. 4. Illustration for RESPA.

C. Reconstructed Module

As pointed out in [45], the reconstruction of the original image information can extract and retain the low-dimensional and high-dimensional information of the image. Inspired by MAE and other state-of-the-art works, two methods of pixel reconstruction are designed for the HSI classification task, i.e., reconstruction based on spatial information and reconstruction based on spectral information for HSI-pixel.

1) Reconstructed Spatial Information Module: As shown in Fig. 4, the output features in the feature extraction module are directly connected to RESPA. In RESPA, the whole patch is masked. A random learnable vector is used to fill all the masked pixels. To increase the difficulty of RESPA, the feature extraction module can better retain all the features of the image, so as to better regularize the classification task. The rate of mask on the feature block is usually selected above 80%.

The feature blocks of the mask are input into the attention module and the traditional convolution module for information extraction and reconstruction, respectively. The purpose is to act as a decoder to restore the original information of the central pixel of the image. After the two postprocessing feature blocks are obtained, the final feature block is obtained by weighted addition using the two learnable parameters. Finally, pooling operation and full connection operation are performed on the feature, so as to obtain the restoration of the original center classification pixels. The MSE is the loss function, and the original pixel tensor is the resulting label in the spatial reconstruction module. The MSE loss used in RESPA is defined as

$$\mathcal{L}_{\text{RESPA}} = \sum_{b=1}^{B} \|\mathbf{y}^b - \mathbf{p}^b\|_2 \tag{16}$$

where $\mathbf{y}^b$ is the original pixel, $\mathbf{p}^b$ is the result of RESPA, and B is the batchsize.

2) Reconstructed Spectral Information Module: As shown in Fig. 1, in addition to the reconstruction module using spatial information, there are also reconstruction modules using spectral information. The feature blocks obtained by the feature extraction module are pooled to obtain a spectral feature that aggregates the information used by the input patch.

The obtained feature spectra were masked, and some points were randomly selected to remove and did not participate in subsequent training. Similarly, the mask rate is usually chosen to be above 80% for increasing the difficulty of RESPE. Then, the feature spectra after the mask were fully connected to obtain the recovery vector of the center pixel corresponding to the input. MSE was used as the loss function, and the original pixel was used as the label for training. The MSE loss used in RESPE is defined as

$$\mathcal{L}_{\text{RESPE}} = \sum_{b=1}^{B} \|\mathbf{y}^b - \mathbf{p}^b\|_2 \tag{17}$$

where $\mathbf{y}^b$ is the original pixel, $\mathbf{p}^b$ is the result of RESPE, and B is the batchsize. Totalizing above loss functions, the final loss of MrDCN is

$$\mathcal{L} = \mathcal{L}_{\text{CLS}} + \mathcal{L}_{\text{RESPA}} + \mathcal{L}_{\text{RESPE}}. \tag{18}$$

D. Analysis on the Proposed Method

In HSI pixel-level classification, spatial features and spectral features represent different information. The spectral information reflects the discriminative information of the material. The spatial information reflects the geometric information of different materials and their distribution information in space. As a result, in feature extraction, the two kinds of information should be decoupled. Therefore, MrDCN adopts DFEM for feature extraction.

For classification tasks, overfitting is a common phenomenon. When the feature extraction module pays too much attention to the high-dimensional features which are beneficial to the current data classification, the phenomenon of overfitting occurs, and a lot of low-dimensional information is lost in the feature extraction process. In fact, this is not good for the generalization of the network. Recently, MAE tasks have proved that reconstruction tasks and training can extract low-dimensional and high-dimensional information of images. However, MAE is a study of visible light. The reconstruction operation can be used as a regularization term instead of pretraining to ensure that the low-dimensional information of the network can always be maintained in the training process of the feature extraction module. Therefore, MrDCN adopts the RESPA module and RESPE module to reconstruct the original spectral information so that MrDCN can retain the discriminative information from low-dimension information and high-dimension information.

Furthermore, the design of the reconstruction module increases the training time and computational complexity, and its network parameters account for a large proportion. However, in the deployment of the real scenes, the network MrDCN can be simplified, the reconstruction module can be removed, and only the feature extraction module and classification module are used. Therefore, in the deployment and use process, the computational complexity of the network derivation is much lower than that of the training process, because the number of parameters is far less than the trained complete network.

III. Experiments and Analysis

In this section, three HSI datasets are used to evaluate the proposed MrDCN's effectiveness. First, three HSI datasets representing various situations—Gaofeng State Owned Forest Farm (GSOFF), Houston2013 (H2013), and Yellow River Estuary (YRE)—are shown. These datasets include the forest,

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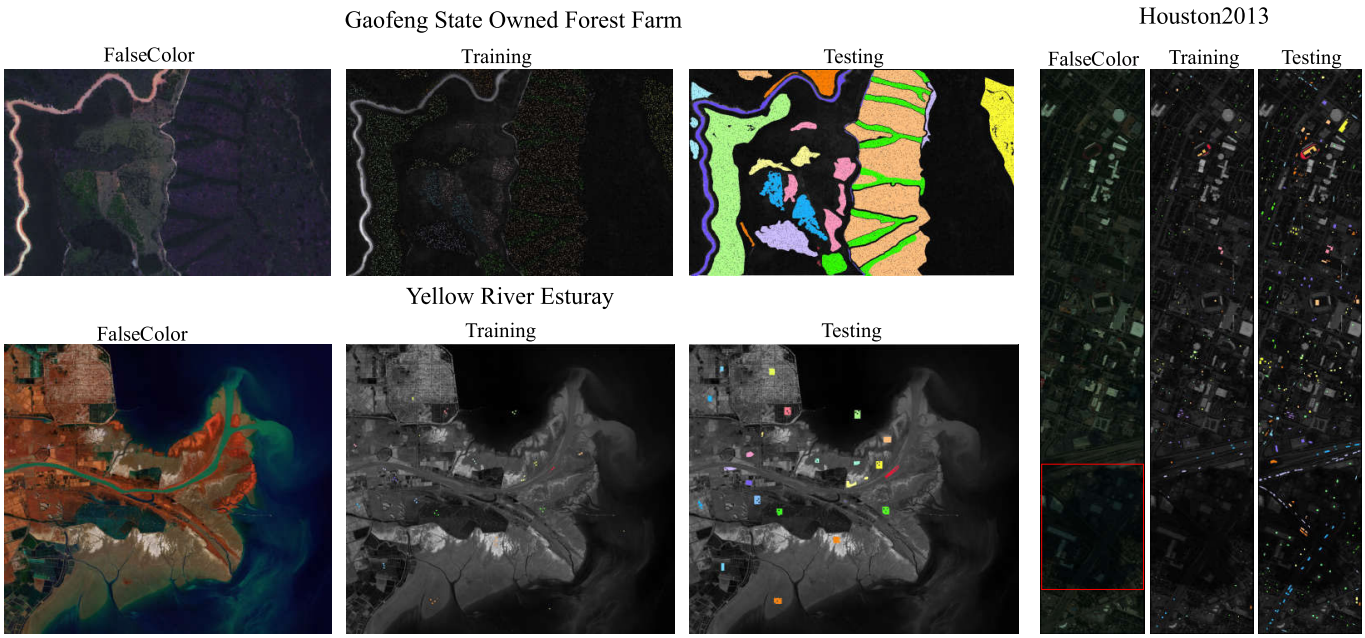


Fig. 5. Maps of false color and training and testing samples' distributions of the GSOFF, H2013, and YRE datasets, respectively.

TABLE I
CLASS LABELS AND TRAIN–TEST SAMPLES' DISTRIBUTION OF GSOFF

Class No.	Class Name	Train	Test
1	Cuninghamia lanceolata	4038	76705
2	Pinus massoniana	268	5072
3	Pius elliottii	208	3967
4	Eucalyptus grandis x urophylla	746	14185
5	Eucalyptus urophylla	2014	38261
6	Catanopsis hystrix	1274	24187
7	Mytilaria laosensis	150	2850
8	Camellia oleifera	416	7908
9	Other broadleaf forest	538	10203
10	Road	524	9948
11	Cutting blank	508	9662
12	Building land	4	64
Total		10688	203012

TABLE II
CLASS LABELS AND TRAIN–TEST SAMPLES' DISTRIBUTION OF H2013

Class No.	Class Name	Train	Test
1	Grass healthy	198	1053
2	Grass stressed	190	1064
3	Grass synthetics	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
Total		2832	12197

urban, and moist soil scenarios, respectively. Second, the ablation tests are conducted to verify the effectiveness of the suggested MrDCN. Then, experiments of MrDCN are conducted in contrast to a number of conventional and state-of-the-art algorithms. Finally, experiments of MrDCN and other algorithms are conducted in embedded device to verify operating efficiency of the suggested MrDCN.

A. Datasets

1) Gaofeng State Owned Forest Farm: In January 2018, over the farm in Guangxi of China, the AISA Eagle II diffraction grating push-broom hyperspectral imager system acquired the dataset which includes 572 × 906 pixels, 1 × 1 m spatial resolution, and 3.3 nm spectral resolution. It contains 125 spectral bands in the wavelength range of 0.4–1.0 μm. It has nine different forest vegetation classes and three other classes. In Table I, the quantity of training and testing samples is displayed. In Fig. 5, the map of false color and sample distribution are shown in Fig. 5.

2) Houston2013: In June 2012, over the University of Houston and neighboring areas, the CASI-1500 sensor obtained this dataset which includes 349 × 1905 pixels. It has 2.5 m spatial resolution and 4.65 nm spectral resolution. It contains 144 spectral bands in the wavelength range of 0.38–1.05 μm. It has ten man-made and five natural objects. In Table II, the quantity of training and testing samples for each class is listed. The map of false color and samples' distribution are shown. Note that in the red region of the map of false color, there is an impact by cloud.

3) Yellow River Estuary: In November 2018, over the Yellow River Delta, GF-5 satellite gathered this dataset which includes 1185 × 1342 pixels. It has 30 m spatial resolution. It has 5 nm spectral resolution and 150 spectral bands in VNIR 0.40–1.00 μm, and 10 nm spectral resolution and 180 spectral bands in the wavelength range of 1.00–2.50 μm in short-wave infrared (SWIR). 1 and 2 of VNIR bands are moved,

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TABLE III
CLASS LABELS AND TRAIN–TEST SAMPLES' DISTRIBUTION OF YRE

Class No.	Class Name	Train	Test
1	Spartina alterniflora	84	743
2	Low-tide mudflat	84	709
3	High-tide mudflat	91	645
4	Mixed area 1	16	123
5	Mixed area 2	95	774
6	Mixed area 3	75	731
7	Oil field	72	612
8	Sea	135	756
9	Intertidal phragmite	86	758
10	Ecological reservoir	75	563
11	Suaeda salsa	67	610
12	Salt fields	81	663
13	Arable land	68	681
14	Phragmite	83	756
15	Woodland	49	502
16	Lotus pond	75	461
17	Typha orientals presl	6	30
18	Robina pseudoacacia	65	516
19	Pond	86	763
20	Yellow river	88	771
Total		1481	12167

TABLE IV
ANALYSIS EXPERIMENTS ABOUT THE SUGGESTED RESPA, RESPE, DFE, AND CAC IN REFERENCE TO THE CLASSIFICATION PERFORMANCE (OVERALL ACCURACY [%]) FOR THE GSOFF, H2013, AND YRE DATASETS

RESPA	RESPE	DFE	CAC	GSOFF	H2013	YRE
×	×	×	✓	84.29	78.16	84.15
✓	×	×	✓	87.98	81.85	88.93
×	✓	×	✓	89.20	82.74	89.82
✓	✓	×	✓	92.68	84.48	94.22
×	×	✓	✓	86.04	80.41	84.77
✓	×	✓	✓	95.93	84.63	91.10
×	✓	✓	✓	95.11	83.38	90.37
✓	✓	✓	×	92.21	83.06	91.42
✓	✓	✓	✓	**97.43**	**87.87**	**96.14**

as well as 175–180, 172–173, 119–121, 96–115, and 42–53 of SWIR spectral bands, leaving 285 bands in total. It has 20 wetland classes. In Table III, the quantity of training and testing samples is displayed. The map of false color and samples' distribution are shown.

B. Ablation Studies

The purpose of the ablation experiment is to validate the proposed MrDCN's reconstruction modules, i.e., RESPA and RESPE. Classification performance is measured using overall accuracy (OA). As shown in Table IV, after RESPA, RESPE and both the modules are added into HSI classification, and there have been major advances in classification performance. In MrDCN, the OA increases 9.89%, 9.07%, and 11.39%, 4.22%, 2.97%, and 7.46%, and 6.33%, 5.6%, and 11.37% for the GSOFF, H2013, and YRE datasets, respectively. Hence, the effectiveness of RESPA and RESPE for HSI classification is confirmed.

To discuss the effect of the DFE module, the traditional convolution is used to replace the DFE module of MrDCN (MrCNN) for analysis experiment, and the effect of the reconstructed module is further discussed. In MrCNN, the OA increases 3.69%, 4.91%, and 8.39%, 3.69%, 4.58%, and 6.32%, and 4.78%, 5.67%, and 10.07% for the GSOFF, H2013, and YRE datasets, respectively. Hence, the effectiveness of RESPA and RESPE is confirmed again.

Comparing the results in Table IV, every term of MrDCN corresponds to a stronger result than MrCNN. Therefore, the feature extraction capability of DFE is stronger than convolution which is the traditional spatial–spectral feature extraction module for HSI. As for the gain brought by the reconstruction module, it is very stable on the YRE dataset, perhaps because the number of bands is large, which can play a good regularization effect in both the methods. However, the number of bands of GSOFF data is the least, so the gain fluctuation of the two methods is large. Therefore, the reconstruction module can retain the discriminative information in the process of feature extraction.

As shown in Table IV, after CAC is added into HSI classification, there have been major advances in classification performance. In MrDCN, the OA increases 5.22%, 4.81%, and 4.72% for the GSOFF, H2013, and YRE datasets, respectively. Hence, the effectiveness of CAC for HSI classification is confirmed.

As shown in Fig. 9, to demonstrate feature extraction performance of reconstruction modules, 512-D features of the trained MrDCN and MrDCN without reconstruction modules are outputted. It is observed that the overlap of other category points in the dataset near the various clustering centers improved, particularly data points in the H2013 data. This phenomenon indicates that the data reconstruction function of the reconstruction module can alleviate the overfitting phenomenon of the network. The results in Fig. 10 illustrate the accuracy of each algorithm under different training sample ratio columns.

TABLE V
1,0,0 ANALYSIS EXPERIMENTS ABOUT THE PARAMETER ma IN REFERENCE TO THE CLASSIFICATION PERFORMANCE (OVERALL ACCURACY [%]) FOR THE GSOFF, H2013, AND YRE DATASETS

ma	0.25	0.5	0.75	0.85	0.9
GOSEF	85.99	93.66	96.21	97.43	97.03
H2013	83.94	83.35	84.07	87.87	84.31
YRE	92.16	92.74	94.23	96.14	95.01

TABLE VI
ANALYSIS EXPERIMENTS ABOUT THE PARAMETER me IN REFERENCE TO THE CLASSIFICATION PERFORMANCE (OVERALL ACCURACY [%]) FOR THE GSOFF, H2013, AND YRE DATASETS

me	0.25	0.5	0.75	0.85	0.9
GOSEF	94.07	93.44.	94.18	97.43	95.37
H2013	82.34	82.11	83.47	87.87	83.25
YRE	93.04	91.43	93.14	96.14	96.03

C. Parameter Analysis

The core parameters for MrDCN are ma and me which are the mask ratio for RESPA and RESPE, respectively. The ma and me are selected from {0.25, 0.5, 0.75, 0.85, 0.9}. As shown in Tables V and VI, with the increase in mask ratio, the difficulty of reconstruction increases, the regularization effect becomes more obvious, and the classification accuracy is

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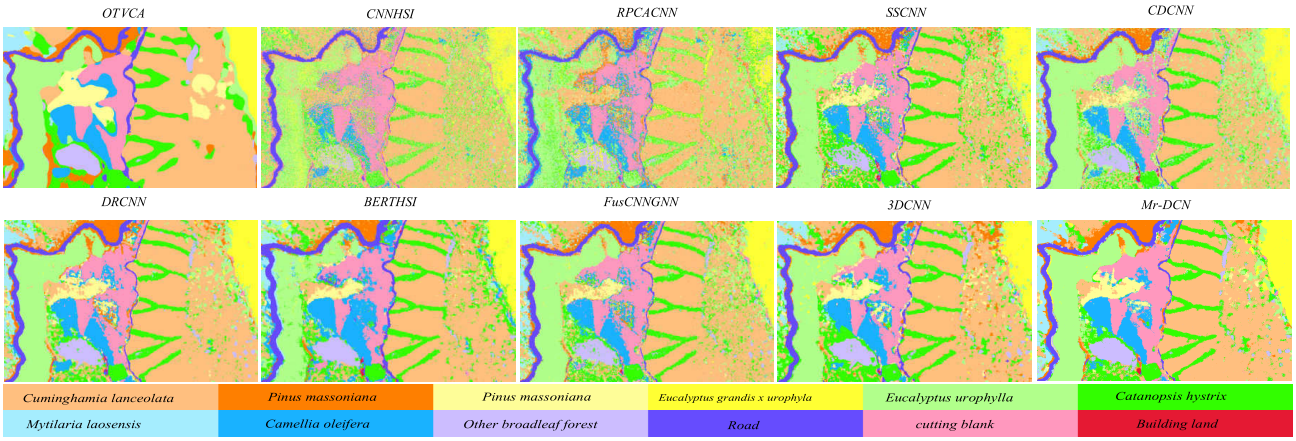


Fig. 6. GSOFF's ground truth and classification maps in different methods.

TABLE VII
CLASSIFICATION RESULTS [%] OF GSOFF WITH DIFFERENT METHODS

Class No.	OTVCA [46]	CNNHSI [42]	RPCACNN [12]	SSCNN [13]	CDCNN [14]	DRCNN [15]	BERTHSI [36]	FusCNNGCN [44]	3DCNN [16]	MrDCN
1	95.34	86.83	83.48	91.61	92.05	**98.00**	96.99	95.78	95.43	97.43
2	**99.70**	11.73	34.27	75.51	77.43	92.61	92.94	89.33	92.84	96.75
3	**99.24**	3.05	20.04	85.53	73.56	92.74	93.80	84.93	97.08	96.49
4	**100**	77.99	52.23	86.07	91.62	99.28	97.46	98.06	96.84	99.68
5	**99.93**	61.86	66.16	96.73	99.09	99.58	92.44	98.48	97.51	99.83
6	83.31	45.59	29.80	85.07	76.52	86.41	90.09	83.82	84.41	**91.12**
7	**99.89**	1.16	6.91	70.28	73.86	96.91	83.58	92.67	94.28	99.52
8	99.03	58.80	72.03	95.52	94.92	99.06	**99.97**	97.77	98.47	99.70
9	91.30	45.53	54.82	68.81	77.59	89.16	93.46	83.40	90.74	**94.40**
10	98.43	97.63	98.13	99.10	99.66	**99.76**	99.69	99.49	99.01	99.59
11	**99.98**	94.78	95.74	99.15	99.22	99.83	98.34	99.81	99.69	99.97
12	0.00	7.81	0.00	**90.63**	65.63	26.56	68.75	48.44	45.31	65.15
OA	95.63	69.59	67.48	90.32	90.60	96.50	95.12	94.42	94.80	**97.43**
AA	88.85	49.40	51.13	87.00	85.09	89.99	92.29	89.33	90.97	**94.97**
Kappa*100	94.48	60.91	58.69	87.79	88.11	95.57	93.84	92.95	93.45	**96.79**

TABLE VIII
CLASSIFICATION RESULTS [%] OF H2013 WITH DIFFERENT METHODS

Class No.	OTVCA [46]	CNNHSI [42]	RPCACNN [12]	SSCNN [13]	CDCNN [14]	DRCNN [15]	BERTHSI [36]	FusCNNGCN [44]	3DCNN [16]	MrDCM
1	80.72	82.53	82.34	82.53	82.43	82.34	83.00	82.62	80.63	**83.09**
2	79.04	83.55	82.33	**84.96**	**84.96**	84.77	83.36	78.10	**84.96**	95.48
3	99.21	99.60	85.35	98.81	**100**	88.71	91.09	91.68	95.84	97.23
4	84.75	**98.58**	91.10	92.23	93.37	92.33	93.56	96.31	71.40	96.21
5	98.20	98.67	93.47	99.24	**100**	99.91	99.81	**100**	90.06	96.59
6	94.41	99.30	83.92	93.01	95.80	99.30	98.60	98.60	**100**	98.60
7	89.93	71.08	74.16	94.12	79.38	89.09	87.22	91.23	**93.56**	82.46
8	49.76	62.58	63.34	68.95	71.23	67.90	68.76	85.85	85.57	**86.99**
9	**85.08**	59.87	76.02	80.64	78.75	78.00	77.15	72.71	72.71	73.75
10	67.37	**96.24**	46.04	43.24	49.71	63.51	48.46	61.49	42.57	68.34
11	74.10	81.69	66.03	59.49	79.22	72.68	82.73	79.51	65.65	**92.13**
12	80.02	61.19	60.42	90.11	78.39	**91.64**	86.17	92.03	78.48	88.86
13	58.25	59.30	78.95	85.61	**92.63**	89.47	86.32	90.53	67.02	88.77
14	**100**	**100**	98.38	95.95	**100**	**100**	99.60	99.60	99.19	**100**
15	97.25	98.10	93.23	94.50	**100**	83.09	98.94	90.70	84.36	**100**
OA	80.62	81.30	75.60	81.63	82.32	83.28	82.90	85.23	78.25	**87.87**
AA	82.54	83.49	78.34	84.23	85.73	85.52	85.65	87.40	80.80	**89.90**
Kappa*100	79.13	79.69	73.66	80.05	80.95	81.94	81.51	83.96	76.49	**86.83**

higher. The accuracy reaches the highest when it reaches 85%. When mask ratio continues to increase, too much information will be lost and cannot be reconstructed, which affects the accuracy.

D. Classification Performance

In Tables VII–IX, the class-specific accuracy (CA), OA, average accuracy (AA), and Kappa coefficient are used to evaluate the classification performance. Besides RPCACNN performing randomized principal component analysis (PCA) before feature extraction that may lose the discriminative information of original data, CNNHSI is the worse method than other spectral–spatial information extraction networks. Because it only extracts spectral information. The H2013 dataset, an urban HSI data, is the most complex in three datasets; hence, 3DCNN performs poorly in the GSOFF and

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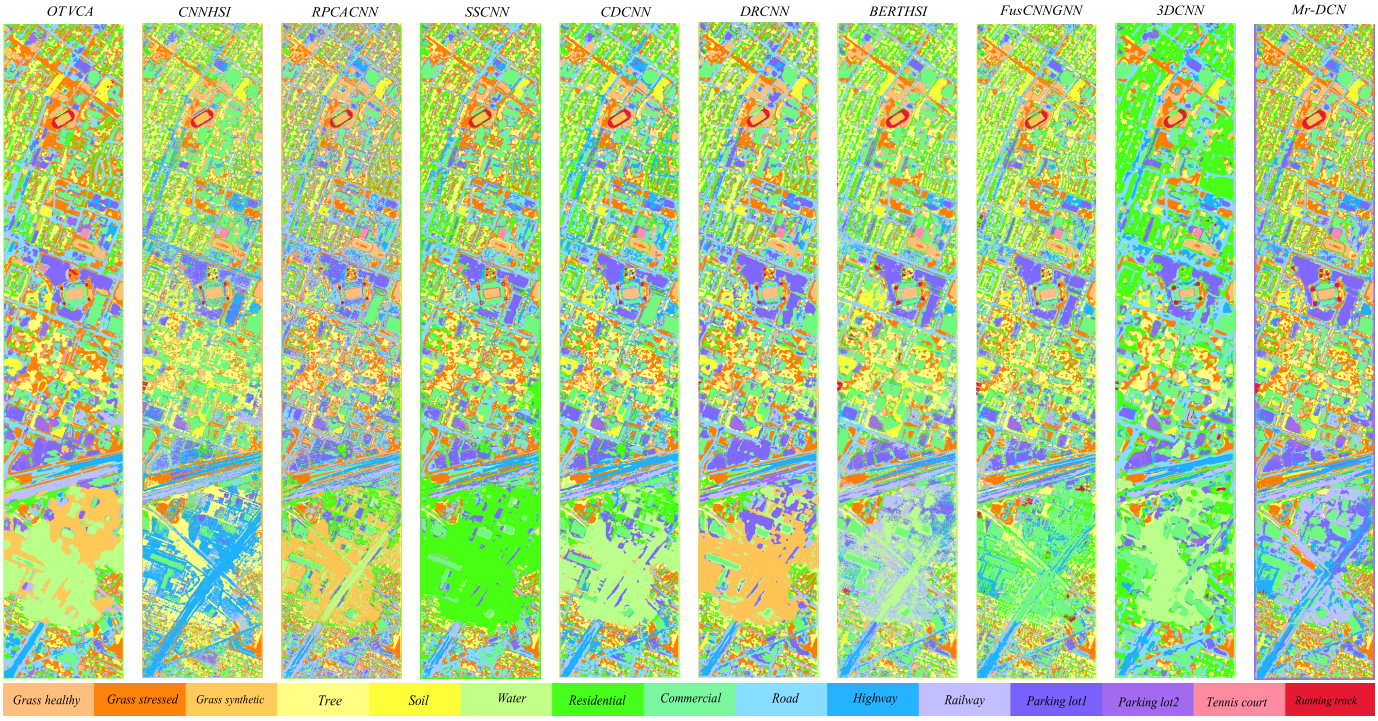


Fig. 7. H2013's ground truth and classification maps in different methods.

TABLE IX
CLASSIFICATION RESULTS [%] OF YRE WITH DIFFERENT METHODS

Class No.	OTVCA [46]	CNNHSI [42]	RPCACNN [12]	SSCNN [13]	CDCNN [14]	DRCNN [15]	BERTHSI [36]	FusCNNGCN [44]	3DCNN [16]	MrDCN
1	100	100	100	100	100	100	100	100	89.91	**100**
2	100	100	100	100	100	100	100	100	100	**100**
3	100	97.52	100	95.97	99.22	100	100	100	100	**100**
4	100	82.93	100	100	100	100	100	100	100	**100**
5	44.32	**75.19**	66.93	85.66	20.41	48.84	43.15	25.58	23.00	50.64
6	**100**	73.60	64.71	81.81	86.05	90.70	**100**	84.27	74.83	99.31
7	100	54.58	46.90	69.61	100	97.71	100	98.37	100	**100**
8	100	100	100	100	100	100	100	100	100	**100**
9	100	99.21	100	100	100	100	100	100	100	**100**
10	100	86.50	78.69	89.52	100	100	100	97.87	100	**100**
11	92.79	96.39	73.11	51.80	91.80	97.38	45.57	42.79	89.02	**100**
12	90.95	63.95	65.91	84.16	70.44	73.30	70.29	62.90	**97.74**	96.08
13	48.90	60.35	47.28	46.99	98.97	91.04	77.97	99.41	89.57	**100**
14	100	96.69	91.80	89.81	100	100	100	100	100	**100**
15	100	79.08	79.88	73.90	86.25	99.60	92.03	82.67	98.80	**100**
16	**95.44**	60.95	30.37	77.22	87.64	86.55	64.64	78.09	84.60	87.64
17	100	93.33	10.00	100	100	100	100	100	100	**100**
18	99.81	81.98	30.43	33.91	0.00	22.09	70.74	62.79	92.05	**100**
19	100	81.65	85.45	97.64	97.90	100	100	100	100	**100**
20	100	100	100	100	100	100	100	100	100	**100**
OA	92.56	84.73	77.90	84.00	86.57	90.15	87.89	86.09	90.75	**96.14**
AA	93.61	84.20	73.57	83.90	86.93	90.36	88.22	86.74	91.98	**96.68**
Kappa*100	92.13	83.83	76.59	83.06	85.78	89.58	87.19	85.27	90.21	**95.12**

YRE datasets in spectral–spatial feature extraction networks. OTVCA, DRCNN, BERTHSI, and FusCNNGCN, however, consistently score well in classification across all the three datasets. The suggested MrDCN performs better and more effectively than all the state-of-the-art techniques listed above by approximately 0.93% in GSOFF, 2.64% in H2013, and 3.58% in YRE, respectively, and surpasses the second-highest other methods in OA.

Especially on the YRE dataset, the results of the proposed MrDCN reflect significant advantages over other algorithms. This may be due to the fact that the number of bands in the data is larger and the spectrum has more information, which increases the difficulty of the reconstruction task and brings a better regularization effect. This may reveal that the classification task is not very difficult for the network, and more consideration should be given to the overfitting problem caused by a single index in the training of the network for classification tasks. Therefore, as a simpler method for extracting spectral–spatial features, MrDCN can obtain better classification results.

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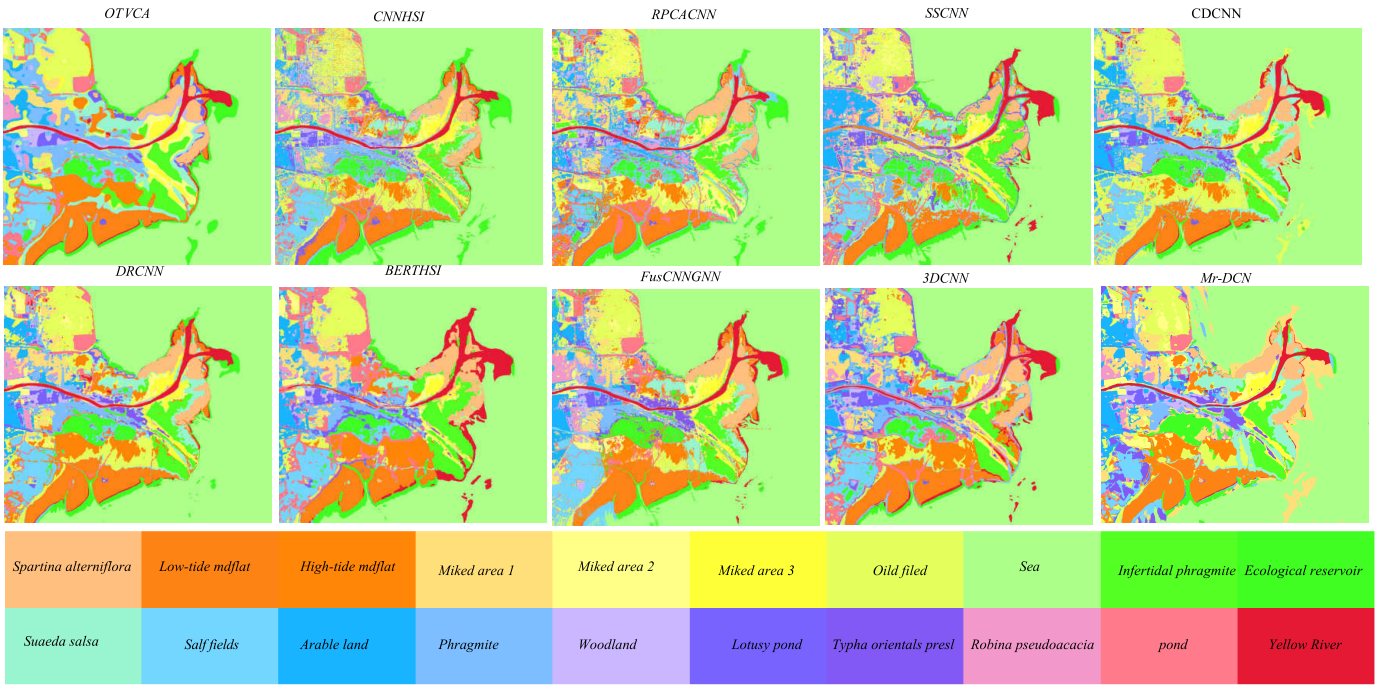


Fig. 8. YRE's ground truth and classification maps in different methods.

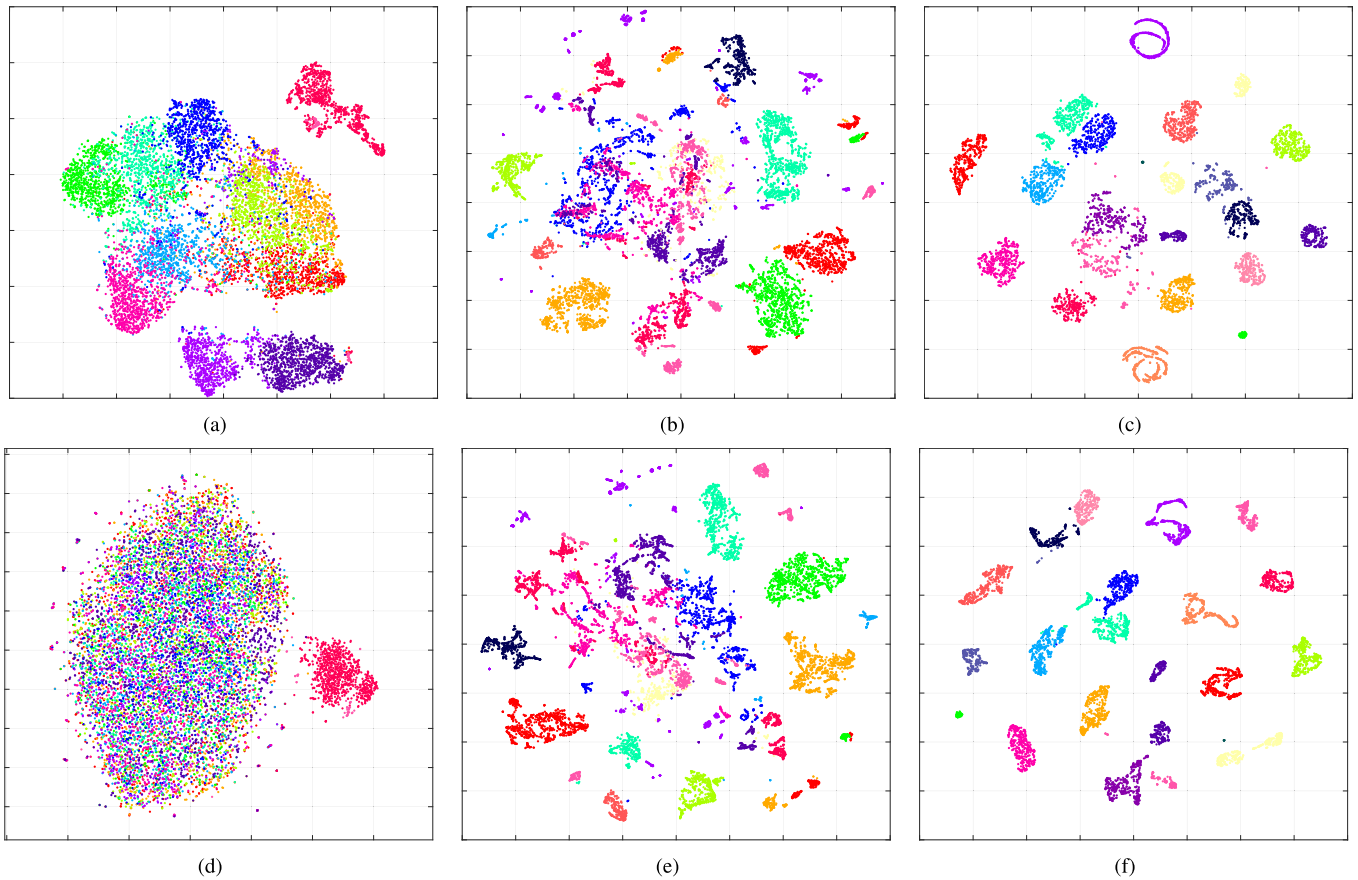
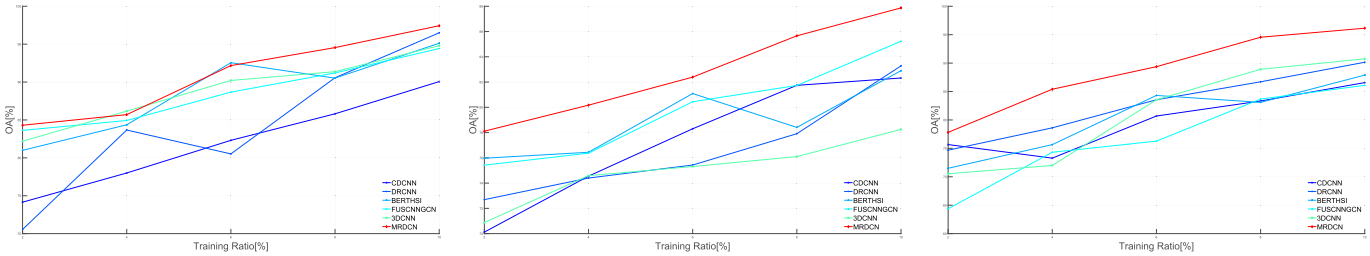


Fig. 9. Feature extraction performance of the proposed MrDCN and MrDCN without reconstruction module using the GSOFF data, H2013 data, and YRE data. (a) GSOFF by MrCDN. (b) H2013 by MrCDN. (c) YRE by MrCDN. (d) GSOFF by MrCDN without reconstruction module. (e) H2013 by MrCDN without reconstruction module. (f) YRE by MrCDN without reconstruction module.

To visually display MrDCN's effectiveness, classification maps of all the methods listed above are displayed in Figs. 6–8. The results of CNNHSI include a lot of salt-and-pepper noise with the worst classification results shown in Tables VII–IX. Because CNNHSI does not use the spatial information. In contrast, even the classification perfor-

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(a) (b) (c)

Fig. 10. Overall classification accuracy of six deep learning methods with better effect versus different percentages of training samples for the three experimental data. (a) GSOFF. (b) H2013. (c) YRE.

TABLE X
INFERENCE TIME OF DIFFERENT METHODS FOR THE GSOFF, H2013, AND YRE DATASETS. UNIT OF TIME IS SECONDS

Method	GSOFF	H2013	YRE
RPCA	2	2	3
CNNHSI	224	300	1068
SSCNN	231	312	1117
BERTHSI	370	538	1579
DRCNN	349	458	1681
fusCNNGCN	401	588	1932
CDCNN	488	642	2028
3DCNN	525	750	3269
OTVCA	3600+	3600+	3600+
MrDCN	340	446	1408

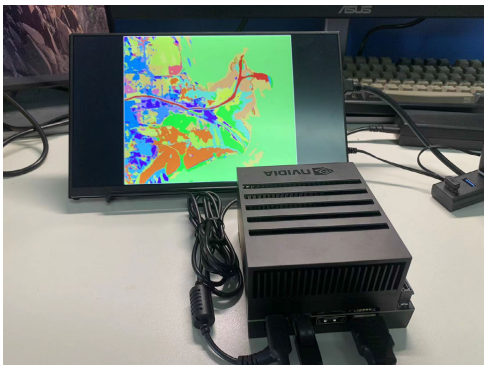


Fig. 11. Experiment of MrDCN for the hyperspectral image in embedded device.

mance of OTVCA is acceptable, the maps are too smoothed to satisfactory, and a lot of minor objects are submerged. However, MrDCN generates classification maps that are more precise, smoother in space, and have less mislabeled pixels than other methods, which is consistent with the outcomes shown in Tables VII–IX. The key challenge with the H2013 dataset is the shadow area covered in clouds, where pixels in the area covered by cloud and other areas exhibit differing spectral properties despite belonging to the same class. Even worse, the majority of training data originate from nonclouded regions. Unsurprisingly, all other cutting-edge methods—aside from CNNHSI—mislabel the highway in the cloud area. However, it is well-classified by MrDCN. This is because MrDCN adds reconstruction modules for regular processing in the process of extracting information, which reduces the appearance of overfitting caused by uneven data distribution. The proposed method extracts the basic properties of pixels in the area covered by cloud and other locations, which can be used to distinguish pixels in the area covered by cloud to some extent using spectral feature and label information.

E. Computational Analysis on Embedded Device

In the actual application scenario, the derivation model can remove the reconstruction module, which greatly reduces the computational complexity and operation time. To illustrate the computational complexities of the proposed MrDCN, using PyTorch in Python on an NVIDIA Jeston AGX Xavier embedded device, Table X offers the inference time of various approaches on the above datasets. The embedded devices are shown in Fig. 11. For all the three datasets, RPCA is the fastest, because it uses PCA preprocessing to unify all the data to the three-channel data and it only includes two simplest convolutions. Therefore, RPCA loses too much information with the worst classification performance. Similarly CNNHSI and SSCNN are the simplest networks, besides the RPCA, which cannot extract enough feature and their classification performance is too bad to satisfy the needs of fine object recognition. OTVCA is a traditional method without GPU, and its time is much higher than 3600 s. Then in the other methods, MrDCN is the fastest method with the best classification performance.

IV. CONCLUSION

In this article, an efficient MrDCN has been suggested for improving the feature extraction and reducing the information loss in HSI classification. In MrDCN, the input HSI patch spectral–spatial information was extracted from the DFEM. At the same time, to reduce the overfitting phenomenon of the feature extraction module caused by the single task of HSI classification, the reconstruction module is used as a branch to train together. To reduce the complexity of MrDCN, the reconstruction module can be removed directly during the derivation. Experiments on the HSI datasets in three different scenarios have demonstrated that the performance of the developed MrDCN outperforms the existing state-of-the-art HSI feature extraction methods, confirming its effectiveness.

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